

Exactness and minimal architectures for differentiable sliced optimal transport plans

Sliced optimal transport can recover not just a distance but an approximate transport *plan*, by lifting the closed-form one-dimensional solution back to the original space along an optimized projection. When this lifted plan *exactly* equals the true optimal transport plan — and how expressive a projector this requires — is not understood. This internship characterizes exactness and the minimal projector architecture that achieves it, validates the predictions numerically, and opens the harder quantitative question of the error away from exactness.

§1. Organization

- **Start date:** As soon as possible (flexible).
- **Localization:** IRISA, Rennes.
- **Supervision:**
 - Romain Tavenard (Prof., Université Rennes 2, MALT Inria team, IRISA, Rennes).
 - Laetitia Chapel (TAGADA research group, IRISA, Vannes).
 - Paul Viallard (MALT Inria, IRISA, Rennes).

§2. Context

Optimal transport (OT) yields both a distance between distributions and an optimal coupling (the transport *plan*) but computing the plan exactly costs $O(n^3)$. Sliced methods reduce the problem to one-dimensional projections, where OT is solved in closed form by sorting and the plan is a monotone permutation. Recent work lifts such a one-dimensional plan back to the original space and *optimizes* the (possibly nonlinear) projector to best approximate the full plan, casting the search as a differentiable bilevel problem (Chapel et al., 2025; Mahey et al., 2023). For this to yield a valid distance, the projection map must be *injective on the support* of the data.

These developments are largely *algorithmic*. The supporting theory, where it exists, concerns the sliced *distance* and says nothing about the *plan*. The most basic question for the plan is still open: when does the lifted, projector-optimized plan *exactly* equal the true OT plan, and how expressive must the projector be for this to be possible? Answering it determines whether the method is a principled approximation or a useful heuristic, tells a practitioner how rich a projector to use, and is the natural first step toward the harder question of how large the gap is when exactness fails.

The core object is the map from a projector to the lifted coupling it induces. The central question is constructive: **for which data, and with how expressive a projector, does this coupling exactly equal an optimal plan?**

This pairs an exactness question with a capacity one. Establishing exactness fixes *which* couplings a class of projectors can realize; the matching question is *how rich* a projector this takes, in terms of minimal width, depth, or number of projections. This is a representation/capacity statement in the spirit of neural-network memorization (Park et al., 2021) and approximation theory (Yarotsky, 2017), but for transport plans rather than function values, and it is open. A *validity* constraint pulls the other way: too rich a projector class breaks the injectivity/identifiability conditions that make the sliced object a valid geometry. The contribution is to locate where the two meet (i.e. find the smallest valid projector that realizes the OT plan) which directly tells a practitioner how to size the projector.

Beyond exactness lies the harder, quantitative question of how the error behaves when the plan is only *approximately* OT. This constitutes an open direction the internship opens, with

neural-network approximation theory (Yarotsky, 2017) as the natural starting point and the optimization-landscape question below as its companion.

§3. Goals

Concretely, the candidate will:

1. formalize the lifted-plan map and a precise notion of exactness that accounts for non-uniqueness of the optimal plan;
2. turn this into a minimal-capacity statement, i.e. find the smallest projector (width, depth, or number of projections) that realizes the OT plan while remaining a valid sliced geometry, drawing on neural-network capacity and approximation results (Park et al., 2021; Yarotsky, 2017);
3. validate the predicted conditions and architectures with controlled experiments on non-trivial geometry, measuring the lifted-plan-to-OT gap;
4. (open direction / Ph.D.) move beyond exact recovery toward quantifying the approximation error, and study the optimization landscape of the projector search so that the solver provably reaches a good projector.

Candidates should possess:

- a strong mathematical background (analysis, linear algebra, convexity; notions of optimal transport are a plus),
- the ability to formulate and prove characterization results,
- good Python coding skills to validate the theory numerically,
- ability to communicate in English in a scientific context.

Funding for a Ph.D. has already been secured: upon a successful internship, the candidate will be offered the opportunity to continue this work as a fully funded doctoral thesis. A successful candidate will develop or reinforce skills key to that continuation, namely:

- a working command of the geometry of optimal transport and sliced approximations,
- experience proving theoretical results and stating them rigorously,
- the habit of pairing theoretical conditions with numerical validation.

§4. Bibliography

- Bonneel, N., Rabin, J., Peyré, G., & Pfister, H. (2015). *Sliced and Radon Wasserstein Barycenters of Measures*. Journal of Mathematical Imaging and Vision.
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- Kolouri, S., Nadjahi, K., Şimşekli, U., Badeau, R., & Rohde, G. K. (2019). *Generalized Sliced Wasserstein Distances*. NeurIPS.
- Mahey, G., Chapel, L., Gasso, G., Bonet, C., & Courty, N. (2023). *Fast Optimal Transport through Sliced Wasserstein Generalized Geodesics*. NeurIPS.
- Park, S., Lee, J., Yun, C., & Shin, J. (2021). *Provable Memorization via Deep Neural Networks using Sub-linear Parameters*. Proceedings of the 34th Annual Conference on Learning Theory (PMLR).
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